RNA-Seq Gene Expression for Cancer Type Prediction

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Goal	Dataset Scale	Best Result
Cancer-type prediction Use gene-expression patterns to classify cancer subtypes and cluster uncertain samples.	**20,531 features** Each patient record contains 20,531 RNA expression values before dimensionality reduction.	**92.5% accuracy** Top hierarchical settings (ward / average) outperformed most alternatives; K-Means reached 91.6%.

Overview

This project investigated cancer type prediction using RNA-Seq gene expression data, with a particular focus on identifying samples from unknown classes. Our research combined ***SVM classification*** with ***K-Means*** and ***Hierarchical Clustering***, allowing low-confidence samples to be further analyzed rather than forced into known categories. The workflow also included ***KNN imputation***, ***PCA-based dimensionality reduction***, and class balancing with ***KMeans-SMOTE*** and ***ADASYN***. Results showed that the proposed hybrid approach improved class separability, with the best clustering configuration achieving ***92.5%*** accuracy.

Methodological Pipeline

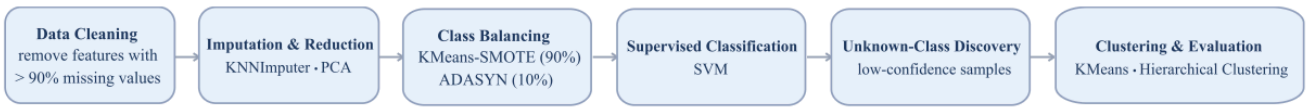

Figure 1. Proposed analytical pipeline for cancer type prediction: preprocessing and rebalancing were followed by SVM classification, while low-confidence samples (confidence < 0.95) were further analyzed using clustering.

1. Data cleaning: The dataset originally contained ***20,531 gene expression features*** with a missing rate of ***14.1%***. By removing features with more than ***90% missing values***, the missing rate was reduced to ***0.4%***, while feature dimensionality decreased by ***22.6%***.

2. Imputation & Dimensionality reduction:

The remaining missing values were completed using ***KNNImputer***. To address the curse of dimensionality, ***PCA*** was applied to reduce the feature space from ***20531*** dimensions to ***292 principal components***, while remaining ***95%*** of the total variance. This resulted in an overall dimensionality reduction of approximately ***98.6%***.

3. Class balancing (Figure 2):

Because the dataset was highly imbalanced across cancer types, a hybrid oversampling strategy was adopted using ***KMeans-SMOTE (90%)*** and ***ADASYN (10%)***. This combination improved class balance while also strengthening samples near difficult decision boundaries.

4. SVM-based classification: ***SVM*** was selected as the primary classifier due to its strong performance on high-dimensional data. In internal validation on known classes, the model demonstrated very strong classification performance, and a confidence threshold of ***0.95*** was applied to identify uncertain predictions.

5. Unknown class discovery: Samples with confidence scores below ***0.95*** were routed to unsupervised clustering for further analysis instead of being directly assigned to known categories. This design allowed the system to better handle potential ***unknown cancer classes*** outside the training set.

6. Clustering and Evaluation: Two clustering methods were compared: ***K-Means*** and ***Hierarchical Clustering***. The results showed that ***K-Means reached 91.6% accuracy***, while ***Hierarchical Clustering achieved up to 92.5%***, though performance varied significantly across linkage strategies.

Post-PCA Class Distribution After Rebalancing

PCA and rebalancing improved class separability. Figure 2 shows clearer class structure after dimensionality reduction and hybrid oversampling, indicating that the preprocessing pipeline enhanced the quality of the feature space.

Hybrid rebalancing strengthened performance. The combination of ***KMeans-SMOTE (90%)*** and ***ADASYN (10%)*** improved class balance while preserving harder-to-learn samples near decision boundaries.

Cluster selection was not based on a single metric. Both ***Silhouette Analysis*** and the ***Elbow Method*** were used to determine cluster counts before comparing clustering accuracy across settings.

Figure 2. Post-PCA class distribution after hybrid rebalancing with KMeans_SMOTE (90%) + ADASYN (10%).

Clustering Performance by Method and Cluster Selection Criteria

Method	Clusters (Silhouette)	Accuracy	Clusters (Elbow)	Accuracy
K-Means	2	91.6%	2	91.6%
Hierarchical - single	2	53.3%	9	53.3%
Hierarchical - complete	2	92.2%	2	92.2%
Hierarchical - average	10	90.4%	6	92.5%
Hierarchical - ward	2	92.5%	2	92.5%

Table 1. Most hierarchical-linkage settings exceeded 90% accuracy, while single-linkage performed substantially worse. K-Means achieved 91.6% accuracy.

Technical Takeaways

Key Insight	Methodological Significance
Clustering performance varied substantially by linkage strategy: hierarchical methods achieved up to ***92.5% accuracy***, whereas single linkage dropped to ***53.3%***.	This project highlighted the importance of **model selection, distance assumptions, and comparative evaluation** in machine learning, especially when different methods respond differently to the same feature space.
This project required handling ***high-dimensional, incomplete, and imbalanced data*** through cleaning, imputation, PCA, and hybrid oversampling before reliable modeling could be achieved.	This experience strengthened my understanding of ***data preprocessing, dimensionality reduction, and robust pipeline design***, which are central to data science and practical machine learning workflows.